Linshan Zhu, Ph.D.

Postdoctoral Scholar, Alfred E. Mann Department of Biomedical Engineering, University of Southern California
Los Angeles, CA | 858-247-8906 | linshanz@usc.edu
ORCID: 0000-0003-3849-3237 | Google Scholar

RESEARCH INTERESTS

Protein engineering and directed evolution for immunotherapy; synthetic receptors and gene circuits; genome engineering for cell therapy; focused ultrasound-controlled cell therapies; live-cell and genome imaging; cancer immunoengineering

EDUCATION AND TRAINING

Postdoctoral Scholar, Alfred E. Mann Department of Biomedical Engineering, University of Southern California — 2023–Present

Ph.D., Bioengineering, Shu Chien–Gene Lay Department of Bioengineering, University of California, San Diego — 2017–2022

B.S., School of Life Sciences, University of Science and Technology of China — 2012–2016

RESEARCH APPOINTMENTS

Postdoctoral Scholar, University of Southern California — 2023–Present

Graduate Student Researcher, University of California, San Diego — 2017–2022

Research Assistant, Shanghai Jiao Tong University — 2016–2017

Undergraduate Researcher, University of Science and Technology of China — 2014–2016

RESEARCH FUNDING AND GRANT ACTIVITY

Ultrasound-Triggered Synthetic Whisper Networks for Selective Immune Regulation — 2026–2028

• Chan Zuckerberg Initiative (CZI); Role: Co-Investigator; PI: Yingxiao Wang; Total budget: $600,000; Status: Funded.

Engineering Ultrasound-Controllable TGF-beta Redirectors to Enhance CAR T Cell Therapy for Glioblastoma — 2026

• NIH/NCI K99/R00 Pathway to Independence Award; Role: PI; Status: A1 resubmitted.

Selected Grant Development Experience

• California Institute for Regenerative Medicine (CIRM) DISC5, 2026: Co-developed with Dr. Lili Yang's group at UCLA; lead writer for the full proposal and coordinator for institutional submission and letters of support.
• CIRM DISC0, 2025: Lead writer for the full proposal; coordinated institutional submission and letters of support.
• CIRM PDEV, 2026: Lead writer for the full translational development proposal; coordinated institutional and external collaborator inputs, supporting documents, and letters of support.
• NIH RPPR and sponsor progress reporting: prepared scientific progress summaries, milestone updates, and data-sharing documentation.

PATENT APPLICATIONS

Synthetic TGF-Beta Redirectors and Uses Thereof in Chimeric Antigen Receptor-Mediated Cell Therapy and Others — Filed 2024

• Application: PCT/US2024/047638; Current status: PCT application entered national phase, including U.S. national phase, assigned to USC.
• Inventors: Yingxiao Wang, Longwei Liu, Linshan Zhu.

HONORS AND AWARDS

Travel Award, UC Systemwide Biomedical Engineering Symposium, Bioengineering Institute of California — 2022

Pass with Distinction, Ph.D. Qualifying Examination, UC San Diego	2018
Seagate Special Scholarship, Seagate Technology LLC	2015
National Motivational Scholarship, Ministry of Education of the People's Republic of China	2013, 2014
Outstanding Volunteer, University of Science and Technology of China	2013
The Scholarship of Freshman, University of Science and Technology of China	2012

MANUSCRIPTS UNDER REVISION

**Co-first author; #Corresponding author.*

1. Liu L.*#, **Zhu L.***, Mouchawar A., Qu Y., Wu Z., Wang Y., Guo T., Chittle S., Nam C., Shao X., He P., Lin D., Liu R., Wang Y.# Reprogramming Immunosuppressive Signals for Sustained Sonogenetic CAR T-cell Therapy. *Cell Stem Cell*. **Under revision.**

2. Zhu L.#, Monther A., Mouchawar A.C., Huang Y., Tsoy M., Yu X., Dong X., Huang Z., Peng Q., Qu Y., Wu Y., Liu L., Wang Y.# Programmable Imaging of Locus-Specific DNA Repair-Associated Dynamics Reveals Chromatin-Controlled Therapy Resistance in Cancer. *Nature Communications*. **Revision submitted.**

SELECTED FIRST OR CO-FIRST AUTHOR PUBLICATIONS

**Co-first author; #Corresponding author.*

1. Zhu L.*, Cui J.*, Man C.W., Mouchawar A., Yu X., Huang Y., Wu Y., Guo T., Xu H., et al. Ultrasound controllable and antigen-independent killing via cis-activating CAR T cells. *Molecular Therapy Oncology*. 2026.

2. Zhu L.*, Man C.W.*, Harrison R., Limsakul P., Peng Q., Hashimoto M., Mamaril A., Xu H., Wu Z., Liu L.#, Wang Y.# Engineering a Programmed Death-Ligand 1-Targeting Monobody Via Directed Evolution for SynNotch-Gated Cell Therapy. *ACS Nano*. 2024.

3. Zhu L., Wu Y., Yoon C.W., Wang Y. Mechanogenetics for cellular engineering and cancer immunotherapy. *Current Opinion in Biotechnology*. 2020.

4. Zamat A.*, **Zhu L.*,** Wang Y. Engineering Molecular Machines for the Control of Cellular Functions for Diagnostics and Therapeutics. *Advanced Functional Materials*. 2019.

ADDITIONAL PEER-REVIEWED PUBLICATIONS

**Co-first author; #Corresponding author.*

1. Yoon C.W., **Zhu L.,** Dong X., Yuan Z., Qu Y., He P., Liu L., Wang Y. Sonogenetics for Precision Medicine: A Focus on Immunoengineering and Genome Engineering. *IEEE Reviews in Biomedical Engineering*. 2026.

2. Guo T., Zhu Z., Qu Y., Yu X., **Zhu L.,** Wang Yu., Liu K., Cui J., Dong X., Yoon C.W., Kilic C., He P., Gou S., Zeng Y., Datta P., Zhou Q., Xu H., Puri I., Liu L., Shen K., Wang Y. Ultrasound Priming Gated by Solid Tumor Hallmarks to Guide CAR-T Therapy, *Science Advances*. 2026.

3. Qu Y.*, Wang Y., **Zhu L.**, Ma F., Yoon C., Chen X., Zhang J., Bishop C., Tang M., Cui J., Guo T., Ho P., Kast M., Xu H., Zhou Q., Engler A., Wang N., Liu L., Chien S., Wang Y. Identifying and reprogramming softness-driven cancer stem-like cells overcomes CAR-T cell resistance in solid tumours. *Nature Biomedical Engineering.* 2026.

4. Yoon C.W.#, Song C., Nguyen D.N.M., Yu X., **Zhu L.,** Ho P., Huang Z., Lu G., Wang Y., Wei F., Qu Y., Zamat A., Lewis A., Chen R., Zeng Y., Datta P., Lee N.S., Jamieson C., Yu B., Shung K., Zhou Q., Liu L.#, Wang Y.#. Tumor Priming by Ultrasound Mechanogenetics for CAR T Therapy. *Nature Materials.* 2025.

5. Limsakul P., Srifa P., Huang Z., **Zhu L.,** Wu Y., Charupanit K. Immunomodulatory Effects of Curcumin on CAR T-Cell Therapy. *Antioxidants (Basel)*. 2025.

6. Liu L., He P., Wang Y., Ma F., Li D., Bai Z., Qu Y., **Zhu L.,** Yoon C.W., Yu X., Huang Y., Liang Z., Zhang Y., Liu K., Guo T., Zeng Y., Zhou Q., Chung H.K., Fan R., Wang Y. Engineering sonogenetic EchoBack-CAR T cells. *Cell.* 2025.

7. Wu Y., Huang Z., Liu Y., He P., Wang Y., Yan L., Wang X., Gao S., Zhou X., Yoon C.W., Sun K., Situ Y., Ho P., Zeng Y., Yuan Z., **Zhu L.,** Zhou Q., Zhao Y., Liu T., Kwong G., Chien S., Liu L., Wang Y. Ultrasound Control of Genomic Regulatory Toolboxes for Cancer Immunotherapy. *Nature Communications.* 2024.

8. Peng Q., Huang Z., Sun K., Liu Y., Yoon C.W., Harrison R., Schmitt D., **Zhu L.,** Wu Y., Tasan I., Zhao H., Zhang J., Zhong S., Chien S., Wang Y. Engineering inducible biomolecular assemblies for genome imaging and manipulation in living cells. *Nature Communications.* 2022.

9. Wu Y., Huang Z., Harrison R., Liu L., **Zhu L.,** Situ Y., Wang Y. Engineering CAR T cells for enhanced efficacy and safety. *APL Bioengineering.* 2022.

10. Wu Y., Liu Y., Huang Z., Wang X., Jin Z., Li J., Limsakul P., **Zhu L.,** Allen M., Pan Y., Bussell R., Jacobson A., Liu T., Chien S., Wang Y. Control of the activity of CAR-T cells within tumours via focused ultrasound. *Nature Biomedical Engineering.* 2021.

11. Pan Y., Yoon S., **Zhu L.,** Wang Y. Acoustic mechanogenetics. *Current Opinion in Biomedical Engineering.* 2018.

CONFERENCE PRESENTATIONS

SBMT 23rd Annual Neurotech Convention, Invited Speaker Apr 2026
Title: Synthetic TGF-β Receptors Rewire and Amplify Sonogenetic CAR-T Cell Therapy

BMES 2025 Annual Meeting, Oral Presentation Oct 2025
Title: Visualizing the Dynamic Locus-specific DNA Repair and Cancer Resistance in Live Cells

BMES 2024 Annual Meeting, Oral Presentation Oct 2024
Title: Engineering a Programmed Death-Ligand 1-Targeting Monobody via Directed Evolution for SynNotch-Gated Cell Therapy

BMES 2023 Annual Meeting, Oral Presentation Oct 2023
Title: Simultaneous Genomic Locus Imaging and Manipulation by Elastin-like Proteins in Live Cells

15th International Conference on Ultrasound Engineering for Biomedical Applications, Oral Presentation Jul 2023
Title: Synthetic TGFβ Redirector Rewiring T-cell Response and Positively Regulating CAR Function in Solid Tumor

BMES 2022 Annual Meeting, Oral Presentation Oct 2022
Title: Simultaneous Genomic Locus Imaging and Manipulation by Elastin-like Proteins in Live Cells

22nd Annual University of California Systemwide Symposium on Bioengineering, Poster Presentation Aug 2022
Title: Simultaneous Genomic Locus Imaging and Manipulation by Elastin-like Proteins in Live Cells

20th Annual UC Systemwide Bioengineering Symposium, Poster Presentation Jun 2019
Title: A temperature-sensitive gene expression cassette in mammalian cells

MENTORING AND TEACHING EXPERIENCE

Research Mentor, Laboratory of Dr. Yingxiao Wang, UC San Diego and USC 2018–Present

- Mentored 3 Ph.D. students, 6 master's students, 3 undergraduate students, and 1 visiting scholar on projects related to synthetic biology, CAR T-cell engineering, and live-cell imaging.
- Trainee outcomes include an NIH F31 fellowship award, a BMES conference presentation, and an undergraduate mentee's matriculation to medical school and later peer-reviewed publication.

Teaching Assistant, University of California, San Diego 2018–2020

- Courses: Molecular Imaging and Quantitation in Living Cells; Molecular Components of Living Systems; Principles of Biomedical Imaging.
- Led discussion sections, held office hours, graded assignments, and prepared remote laboratory materials during the COVID-19 pandemic.

Guest Lecturer, University of Southern California 2026

- BME599: Current Trends and Developments in Biomedical Engineering.

PROFESSIONAL SERVICE

- Manuscript reviewer: Cellular Immunology; Frontiers in Bioengineering and Biotechnology; Scientific Reports; Frontiers in Immunology.
- Judge, USC Graduate Grodins Symposium, 2024 and 2026.
- Judge, 76th Los Angeles County Science and Engineering Fair, 2026.
- Session Chair, International Symposium of Genetic Engineering for Cell Imaging and Therapy (ISGECIT), Los Angeles, CA, 2025.